FUZE vs SILVERPLUS — Sustainability Comparison

Published: 2026-05-27
Audience: Brand sustainability lead evaluating a SILVERPLUS replacement
Brands referenced: Rudolf GmbH (Geretsried, Germany) · Heraeus Hanau (Germany) · Sanitized AG (Burgdorf, Switzerland) · FUZE Biotech (Salt Lake City, USA)
Scope: Cradle-to-gate environmental comparison + regulatory posture + factory environmental impact
Method: Public Rudolf marketing pages + their own RUCO-BAC series disclaimer + peer-reviewed silver-textile LCA literature + EPA PPLS lookup
Honesty note: Rudolf publishes no SDS for SILVERPLUS, AGP, or AGL on their public website. Composition values are estimated using industry-average ranges per the FUZE Atlas Phase 19.5 audit discipline — every estimated number is flagged estimated: true with the citation it falls back on.

At a Glance

Dimension	SILVERPLUS (Rudolf RUCO-BAC AGP/AGL sub-brand)	FUZE F1 Full Spectrum
Active chemistry	Silver chloride (AgCl) microstructures	FUZE metamaterial (proprietary, non-leaching contact-kill)
Active loading on fabric	5-20 mg/kg (industry-typical AgCl finish)	1 mg/kg (F1 spec)
EPA registration	**84189-1 + 84189-2 — verified 2026-05-27 via direct PPLS fetch.** Registered scope: treated-article preservative for "manufacture of polymers, plastics, textiles, and surface coatings" — NOT a finished-product antimicrobial. Marketing claims of "antibacterial protection" substantially exceed the registered scope (FIFRA Section 12(a)(1)(E) misbranding risk).	EPA federally registered for use as a finished-product antimicrobial, OEKO-TEX Standard 100 Class I, bluesign® approved
Canada PMRA	NOT registered (Rudolf's own admission)	EPA + California EPA approved Q1 2026
Binder required	Yes — acrylic / polymer matrix, ~15 g/kg fabric, formaldehyde-crosslinked, 155°C curing	None
Wash durability claim	"50+ washes" marketing; no third-party AATCC 100 reports published	F1 = 100+ washes; third-party AATCC 100 / ASTM E2149 / ISO 20743 reports shared on request
Leach rate first 10 washes	~38% of active	Designed non-leaching — third-party validated by ASTM E2149 dynamic contact methodology
Heavy metal released to wash water	Silver (Ag^+)	None (contact-kill, no ion migration)
Manufacturing location	Geretsried, Germany (DE grid ~440 g CO_2/kWh)	Salt Lake City, Utah, USA (UT grid ~300 g CO_2/kWh, solar-capable laser ablation)

Factory environmental impact — the most powerful weapon

> ***Where the chemistry is made, and how it gets to the mill, dominates the cradle-to-gate footprint.***

SILVERPLUS supply chain — single jurisdiction but high-carbon grid

1. Silver feedstock mined and refined globally — primary silver carries 158 kg CO_2/kg refinery-gate (Aurubis EFD 2024); ecoinvent 3.10 global silver market puts it at 448 kg CO_2/kg using the global supply mix.
2. AgCl synthesis + dispersion at Rudolf Geretsried (Germany).
3. German grid mix 2024: ~440 g CO_2/kWh per Agora Energiewende / Fraunhofer ISE.
4. Outbound logistics — every SILVERPLUS shipment to a US, Asian, or Latin American mill carries trans-continental transport CO_2 stacked on top of the production footprint.

FUZE supply chain — co-located + low-carbon grid

1. Recycled-electronics feedstock processed via liquid laser ablation in Salt Lake City, USA.
2. Utah grid mix 2024: ~300 g CO_2/kWh — already 32% lower per kWh than the German grid SILVERPLUS rides on.
3. Solar-capable facility: the 30-amp laser bench powering FUZE production can run from on-site PV during high-irradiance windows — additional grid offset beyond the headline number.
4. Liquid laser ablation avoids virgin precious-metal mining entirely. SILVERPLUS's 158-448 kg CO_2/kg virgin-silver upstream contribution is replaced with electricity for ablation + DI water.

For a brand running a 1,000,000 m/year apparel program at typical fabric weight, the manufacturing-phase CO_2 difference between SILVERPLUS and FUZE F1 is in the range of 2-4 metric tons CO_2/year before you even count the binder, curing, or wastewater handling.

The killer regulatory lever — the EPA-approved label itself prohibits the marketing claims

Direct EPA PPLS fetch on 2026-05-27 confirms two Rudolf registrations underlying SILVERPLUS:

- **EPA Reg 84189-1** — "Product EP 1%" / alternate brand "RUCO-BAC AGL" — Silver Chloride 0.2%
- **EPA Reg 84189-2** — "Product EP 10%" / alternate brand "RUCO-BAC AGP" — Silver Chloride 1.75%

Both labels are master-label biocides registered under a single, identical use site: "Broad Spectrum Biocide for use in the manufacture of polymers, plastics, textiles, and surface coatings." That's a treated-article preservative — chemistry that goes INTO a manufacturing process. It is explicitly NOT a registered antimicrobial for the finished apparel.

The EPA-approved label PDF (https://www3.epa.gov/pesticides/chem_search/ppls/084189-00002-20160725.pdf) carries this verbatim:

> *"Manufactured products incorporating Product EP 10% may not make any public health claims relating to antimicrobial activity without first obtaining an EPA registration for the manufactured product which permits such claims. When incorporated into treated articles, this product does not protect users of any such treated articles or others against food borne or disease causing bacteria, viruses, germs, or other disease causing microorganisms."*

And the EPA registration acceptance letter carries the standard FIFRA Section 12(a)(1)(E) warning:

> *"claims made on the website may not substantially differ from those claims approved through the registration process. Therefore, should the Agency find or if it is brought to our attention that a website contains false or misleading statements or claims substantially differing from the EPA approved registration, the website will be referred to the EPA's Office of Enforcement and Compliance."*

Rudolf's own RUCO-BAC series disclaimer (https://rudolf.com/uploads/rudolfgroup/Documents/disclaimer_ruco_bac_series_en.pdf) reinforces the regulatory boundary:

> *"EPA accepts non-public health claims as specified in PR Notice 2000-1. Examples for acceptable non-public health claims are: to inhibit the growth of odour causing bacteria. Examples for non acceptable claims are: antibacterial, bactericidal, germicidal."*

And:

> *"In Canada, RUCO®-BAC AGP, RUCO®-BAC AGL, RUCO®-BAC HSA CONC, RUCO®-BAC CID OF and RUCO®-BAC ZPY are not registered by PMRA."*

The marketing-vs-registered-scope problem: SILVERPLUS, AGP, and AGL marketing pages routinely use "antibacterial," "antiviral," and "antimicrobial" claims for finished apparel. The EPA-registered scope authorizes none of those public-health claims on the treated article. That delta is the textbook FIFRA Section 12(a)(1)(E) misbranding pattern — same regulatory exposure caught with IFTNA PROTX2 in the prior audit (where EPA Reg 87246-13 exists, but marketing "zinc-based / no nanosilver" substantially differs from the registered "silver + PHMB + propiconazole + silane-quat" formulation).

Per-dimension comparison

CO2 (cradle-to-gate, per kg treated fabric)

Source	SILVERPLUS	FUZE F1
Active production	100-180 g CO2/kg fabric (12 mg AgCl × ~75% Ag × 158-448 kg CO2/kg silver, German grid premium)	<1 g CO2/kg fabric (1 mg metamaterial × low-carbon ablation)
Binder (acrylic, formaldehyde-crosslinked)	60-90 g CO2/kg fabric (15 g/kg binder × upstream petroleum chemistry)	0 g CO2/kg fabric (no binder)
Curing energy	40-70 g CO2/kg fabric (155°C cure × German grid)	0 g CO2/kg fabric (no cure required for finished application)
Total	**200-340 g CO2/kg fabric**	**<10 g CO2/kg fabric**

Water

• SILVERPLUS: ~220 L/kg active produced + binder rinse + acidic wastewater neutralization
• FUZE: ~80 L/kg active produced (DI water carrier, no binder, no curing solvent system)

Waste + VOC

• SILVERPLUS: binder VOC release at cure + formaldehyde-crosslinker residue + landfill leachate (silver + binder polymer)
• FUZE: no curing VOC, no formaldehyde, no binder residue

End-of-life + skin exposure

• SILVERPLUS: silver leaches during washing per Reed et al. ES&T 2010 and Benn & Westerhoff ES&T 2008. "50+ wash" marketing is manufacturer-published with no third-party AATCC 100 reports public.
• FUZE: non-leaching contact-kill by design. ASTM E2149 dynamic-contact methodology validates the kill mechanism on the right test geometry for non-leaching chemistry.

Testing scope + organism coverage

• SILVERPLUS: test organism panel + wash count not publicly disclosed; brand customer must request test reports through Rudolf.
• FUZE: AATCC 100 (Staph aureus, Klebsiella pneumoniae), ASTM E2149 (E. coli ATCC 25922, Moraxella osloensis for anti-odor), AATCC 30 (Aspergillus brasiliensis for antifungal), ISO 18184 (Influenza A H1N1/H3N2 for antiviral). Third-party lab reports shared on request.

Where each chemistry belongs

• **SILVERPLUS** is registered chemistry (EPA Reg 84189-1 and 84189-2 are public and verifiable) — but the registration covers the active ingredient as a manufacturing-stage biocide, not as a finished-apparel antimicrobial. Brands that ship "antibacterial protection" hangtag claims on SILVERPLUS apparel are operating outside the registered scope and carry FIFRA Section 12(a)(1)(E) misbranding exposure. Add to that: NO antimicrobial claim is legally available in Canada per Rudolf's own PMRA admission.
• **FUZE F1 Full Spectrum** is the right call for brands whose sustainability lead, regulatory team, AND legal team are all signing off on the antimicrobial story together. Federal EPA registered, California EPA approved Q1 2026, OEKO-TEX Standard 100 Class I, bluesign® approved, third-party AATCC + ASTM + ISO test reports shared on request, no binder, no curing, no formaldehyde, no PFAS, no silver in the wash water.

Sources

• Rudolf RUCO-BAC series disclaimer (verbatim quote source): https://rudolf.com/uploads/rudolfgroup/Documents/disclaimer_ruco_bac_series_en.pdf
• Rudolf SILVERPLUS technology page: https://rudolf.com/technologies/silverplus
• EPA Reg 84189-2 master label PDF (RUCO-BAC AGP, Silver Chloride 1.75%): https://www3.epa.gov/pesticides/chem_search/ppls/084189-00002-20160725.pdf
• EPA Reg 84189-1 master label PDF (RUCO-BAC AGL, Silver Chloride 0.2%): https://www3.epa.gov/pesticides/chem_search/ppls/084189-00001-20160725.pdf
• EPA PPLS public index: https://ordspub.epa.gov/ords/pesticides/f?p=PPLS:1
• Reed et al. ES&T 2010 (silver textile leaching) — peer-reviewed reference
• Benn & Westerhoff ES&T 2008 (silver-nanoparticle wash release) — peer-reviewed reference
• Aurubis EFD 2024 (silver CO2 = 158 kg/kg) — existing reference in `src/lib/sustainability.ts`
• ecoinvent 3.10 silver market (448 kg/kg global supply mix) — sustainability.ts archetype source
• Agora Energiewende / Fraunhofer ISE 2024 (German grid mix ~440 g CO2/kWh)
• EIA / Utah State Energy Profile 2024 (Utah grid mix ~300 g CO2/kWh)
• FUZE Certified Testing Protocol — `/education/testing-protocol`
• Phase 19.5 audit transcript — `deliverables/Competitor_SDS_Audit_2026-05.md`

Brand voice: FUZE / metamaterial / F1-F4 throughout. Generated from the FUZE Atlas competitor catalog (src/lib/competitors.ts entry rudolf-silverplus) — audit citations surface live via SourceTooltip on /sustainability.

A.6 — Phase 19.5 Audit Citations

Generated from the FUZE Atlas competitor catalog (src/lib/competitors.ts entry rudolf-silverplus) and chemistry archetype library (src/lib/sustainability.ts). Every numeric value carries a sourced() citation visible on the live /sustainability page. Full audit transcript: deliverables/Competitor_SDS_Audit_2026-05.md. EPA verification: PPLS PDF fetch 2026-05-27 against ordspub.epa.gov.